

Philip Khor

DATA SCIENTIST

Penang, Malaysia

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Summary

Economist by training, data scientist by practice. Python and R across ML modelling, LLM pipelines, and applied econometrics – ESG/sustainability, regulatory machine learning, and economic policy. Technical educator across industries – banking, think-tanks, and corporate training.

Skills: Python, R, SQL • XGBoost (monotonic constraints), scikit-learn • LLM/agent systems: prompt architecture, RAG, Flowise • Agentic coding: Claude Code, OpenCode • MongoDB, DuckDB, AWS • Tableau • Credit risk: Basel, IFRS 9 • ESG/ISSB: FTSE Russell methodology, IFRS S1/S2

Work Experience

Technical Consultant

Remote

SUSTENYX

Mar 2026 - May 2026

- Shipped full-stack PRs (frontend and backend) across a TypeScript-based MERN-stack codebase using agentic coding tools (Claude Code, OpenCode)
- Refactored core modules toward an OOP architecture to improve maintainability ahead of handover
- Led code review and developer knowledge transfer for continuity post-departure

Sustainability Intelligence & Compliance Manager

Remote

SUSTENYX

June 2025 - Feb 2026

- Translated FTSE Russell ratings methodology and IFRS Sustainability Disclosure Standards into AI-powered intelligence systems and commercial offerings for corporate issuers in Malaysia
- Architected a multi-stage LLM prompt-engineering instruction set for ESG disclosure analysis, built on a config-driven (YAML) framework decoupling the entire pipeline logic from application code
- Evaluated and selected document parsing/OCR and LLM provider infrastructure for production deployment across accuracy, cost, and structured-output reliability
- Built a manual ground-truth evaluation set against ISSB disclosure criteria to catch systematic model error patterns and fed corrections back into prompt design
- Drove cost conversations alongside LLM workflow development – surfacing cost-performance trade-offs for model selection, caching architecture, and cost-per-job tracking
- Remediated infrastructure security gaps, bringing AWS environment compliance from 70% to 92%

External Trainer

Remote

FORWARD COLLEGE

Late 2025

- Designed and delivered two workshops on building agent workflows with Flowise for technical staff and subject matter experts

Sustainable Finance Consultant (concurrent, part-time)

Remote

SUSTENYX

Aug 2024 - June 2025

- Conducted in-depth research on ESG ratings, standards and frameworks, and competitor analysis
- Developed reports, presentations, and thought leadership pieces on key sustainability topics for internal and external audiences
- Utilized data analysis and AI tools to refine ESG product development and recommendation models

Visiting Data Scientist

George Town, Penang

PENANG INSTITUTE

Jan 2024 - Dec 2024

- Designed and developed over 12 interactive Tableau dashboards covering population, labour, healthcare, housing, and state finances
- Streamlined data pipelines to reduce update times
- Organised internal workshops on using large language models in policy research

Statistician, Socioeconomics and Statistics Programme

George Town, Penang

PENANG INSTITUTE

Jan 2023 - Dec 2023

- Developed socioeconomic dashboards and maintained data pipelines
- Supported the transition of Penang Monthly's statistics brief toward a more visual format
- Contributed to Penang Monthly statistics articles and prepared research reports

Manager, Regional Advanced Risk Modelling, Group Risk

Kuala Lumpur

CIMB BANK BERHAD

March 2020 - May 2022

- Reviewed and enhanced credit risk model development processes for Basel and IFRS 9 across ASEAN markets
- Defined the regulatory strategy for the team's machine learning credit screening (A/B) models, scoping model development toward BNM regulatory acceptance with IFRS 9 provisioning as the end use case
- Benchmarked model governance and fairness documentation against MAS' FEAT principles and tracked the model-risk stance of other supervisors (e.g. ECB) to anticipate where BNM requirements were heading
- Communicated model outcomes to stakeholders and prepared documentation for machine learning models
- Worked with model validators and data engineers on feature derivation
- Developed R scripts and internal packages to automate analysis on the AutoML platform
- Upskilled risk modellers on ML algorithms, model explainability, and fairness considerations for credit risk ratings models

Technical Writer

George Town, Penang

SKYMIND AI BERHAD

October 2019 - December 2019

- Prepared documentation on internally developed model deployment software to support internal and external users.
- Provided editing assistance for development of training materials.
- Drove and provided support for community engagement events.

Data Scientist Intern

Minato-ku, Tokyo

GRID INC.

May 2019 - August 2019

- Trained real-time object detection and image segmentation models for use in agriculture using YOLOv2, SSD and Mask R-CNN deep learning architectures with internally built framework.
- Delivered substantial improvement on the accuracy of number of trees counted, relative to previously built internal model.
- Developed metrics to evaluate model to align with client objectives.
- Developed pipeline consisting of Python command line interfaces to facilitate pre-processing, partitioning and post-processing of image and annotation data.

Data Scientist Associate

Bangsar South, Kuala Lumpur

THE CENTER OF APPLIED DATA SCIENCE

July 2018 - May 2019

- Delivered over 100 hours in training sessions for Microsoft Excel, R for Analytics, Tableau, Statistics and Data Storytelling.
- Developed training products in the areas of statistics, exploratory analysis, data visualization and machine learning using spreadsheets, R, Python and Tableau delivered through tools such as Jupyter Notebooks and R Markdown, focusing on commercial applications of data science.
- Led development teams for Excel Analytics and Regression Modelling and Evaluation modules.
- Core developer of introductory and intermediate Statistics and R for Analytics modules.
- Worked with teams to develop and deliver training material for Natural Language Processing, Neural Networks, Data Visualisation with Tableau, A/B/n Testing and Time Series Analytics and Forecasting modules.
- Developed internal conventions for material formatting.

Research Assistant: Economics Studies

George Town, Penang

PENANG INSTITUTE

Dec 2015 - Jan 2016, Aug 2016 - Jan 2017

- Provided key research assistance for the Penang Skilled Workforce Study project.
- Scheduled and performed interviews with Human Resource, company executives and other governmental and non-governmental stakeholders.
- Prepared reports and edited report text and formatting.
- Emcee at the forum Budgeting Priorities in a Challenging Economic Environment (Nov 2016).
- Contributed to the Penang Economic and Development Report 2015/2016 and the Institute's evaluation of the Trans-Pacific Partnership Agreement (TPPA).

Research & Projects

Key Penang Statistics

Penang Institute

2023 - 2024

- 12 dashboards providing numerical insights into key socioeconomic issues in Penang for public consumption
- <https://statistics.penanginstitute.com>

Penang Strategy for Economic Ecosystem Development

Penang Institute

2023

- Design visualisations of key economic indicators at sector and aggregate level
- Develop policy recommendations adapting various policy documents to local context

Estimating Penang's Economic Growth

Penang Institute

2023

- Developing regional input-output models to estimate effects of economic shocks to Penang's GDP
- Re-implemented Excel model in Python
- Assisted with acquisition of local data

Penang Economic and Development Report 2022/23

Penang Institute

2023

- Prepared writeups for water and electricity, global business services, waste management, housing and climate change in Penang
- Assisted with acquisition of data and editing

Family background and child health: the role of socioeconomic status and skills University of Wollongong

HONOURS THESIS

2017

- Econometric analysis of the relationship between parental income and child health status in Australia using two large Australian longitudinal datasets.
- Used data from the Household, Income and Labour Dynamics in Australia Survey (HILDA) and the Longitudinal Study of Australian Children (LSAC) to replicate prior longitudinal results on the effect of parental income on child health.
- Developed a novel identification strategy to estimate the effect of parental resources on child health.

Penang skilled workforce study: labour skills for growth and change

Penang Institute

Aug 2016 - Jan 2017

- Identification of skill gaps and skill shortages in major Penang economic sectors.
- Conducted and assisted with qualitative interviews with industry stakeholders and human resource personnel.
- Assisted in the development of questionnaires for data collection and strategies to improve response rates.
- Assisted in data analysis and designing data visualizations.

Selected articles and reports:

- *Penang's semiconductor crossroads: navigating global trends and industrial upgrading*. Penang Monthly, Jan. 2025.
- *Penang's evolving schooling preferences*. Penang Monthly, Aug. 2024.
- *US-China tensions and the transformation of Penang's tech industry*. Penang Monthly, Oct. 2024.
- *The Limited Local Benefits of Swiftonomics*. Penang Monthly, Apr. 2024 [Feature article].
- *EPF Account 3: Flexibility or Erosion of Retirement Savings?*. Penang Monthly, Jun. 2024.
- *Non-Profit Organisations In Penang*. Penang Monthly, Jun. 2024.
- *Families on the edge, Penang: mixed methods research on the impact of COVID-19 and the rising cost of living on children and women in low-income families in Penang* (with M. T. Beh, P. J. Yeong). UNICEF Malaysia, Apr. 2024.
- *When Less Is More Expensive: The Trend of Shrinkflation*. Penang Monthly, Sep. 2023.
- *Artificial Intelligence In Numbers*. Penang Monthly, Nov. 2023.
- *The Jetties of George Town* (with Azmil Akil). Penang Monthly, Jul. 2023.
- *Penang Talent Prospects Survey 2023: Surging Demand for Engineering and IT Talent* (with W. L. Ong). Penang Institute, 2023.
- *Sports and Recreation Statistics* (with Azmil Akil). Penang Monthly, May 2023.
- *On Libraries, Printers and Publishers*. Penang Monthly, Aug. 2023.
- *Expenditure On the Arts: An Overview*. Penang Monthly, Apr. 2023.

Education

Bachelor of Commerce (Honours)

NSW, Australia

UNIVERSITY OF WOLLONGONG

2017

- Received the 2017 Honours Scholarship from the Faculty of Business based on academic performance, research and leadership potential.
- Undertook a research project as part of coursework for Advanced Macroeconomic Theory (Health and wealth: economic growth with endogenous health and human capital).

Bachelor of Commerce (Dean's Scholar)

NSW, Australia

UNIVERSITY OF WOLLONGONG

2014-2016

- Awarded the Illawarra Regional Information Services John Steinke Prize in Economics (April 2017) for achieving the highest mark in 3rd-year Economics subjects.
- Awarded the Kenneth A. Blakey Prize (March 2015) for achieving the highest aggregate mark in Microeconomic Theory and Policy.
- Consistently placed in the Dean's Merit List (2014-2017), which admits the top 5% of students in the Faculty of Business.
- Awarded the Business International Academic Merit Scholarship and the Undergraduate Excellence Scholarship in recognition of academic achievement.
- Admitted to the Dean's Scholar program in recognition of high academic achievement.

Diploma in Business Studies

George Town, Penang

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT & TECHNOLOGY (FORMERLY TUNKU ABDUL RAHMAN UNIVERSITY COLLEGE)

2012-2014

- Received the Merit Scholarship amounting to 100% of the tuition fees for the length of the program in recognition of prior academic performance.
- Awarded Book Prize for overall academic performance.

Qualifications

Bank Risk Management

Kuala Lumpur

ASIAN INSTITUTE OF CHARTERED BANKERS

2022

- Provide banking professionals who aspire to be risk specialists with a comprehensive understanding of bank risk management activities.

Data Scientist Trainee

Bangsar South, Kuala Lumpur

ASEAN DATA ANALYTICS EXCHANGE

2018

- Data science finishing school for graduates covering R and Python programming modules, statistics, data preprocessing, machine learning, data visualisation with programming packages (ggplot2, matplotlib, seaborn) and Tableau, forecasting and natural language processing.